

PARENTERAL ANTICOAGULANTS

VENOUS THROMBOEMBOLISM

Prepared by PRIC Hosp. Sg Buloh & Wong Mun Yen

DOSING REGIMEN

ENOXAPARIN^{1,2}

Prophylaxis:

Standard: 40mg OD

BMI $\geq$ 40kg/m²: 40mg BD

CrCl < 30ml/min or HD: 20mg OD

Treatment:

Standard: 1mg/kg BD

Obesity: 0.75mg/kg BD (actual body weight)

CrCl < 30ml/min: 1mg/kg OD

FONDAPARINUX¹

Prophylaxis:

< 50kg: Contraindicated

> 50kg: 2.5mg OD

Treatment:

< 50kg: 5mg OD

50-100kg: 7.5mg OD

> 100kg: 10mg OD

Contraindicated in CrCl < 30ml/min, pregnant & breastfeeding women.

HEPARIN^{1,3}

Prophylaxis (SC):

Standard: 5,000 units BD or TDS

BMI $\geq$ 40kg/m²: 7,500 units BD

Treatment:**Option A**

(For underweight, overweight, elderly)

Initial IV slow bolus:

(Max 10,000 units/dose)

age < 70: 80 units/kg

age $\geq$ 70: 50 units/kg

BMI $\geq$ 40kg/m²:

Use adjusted body weight

Followed by IV infusion:

18 units/kg/hr (Max 2,000 units/hr)

*Refer to **HSgB IVI Heparin Protocol** for details on APTT monitoring, dose adjustment & protamine antidote.*

Option B

(For patients not specified under Option A)

Initial IV slow bolus:

5,000 units

Followed by IV continuous infusion:

1,000 units/hr

Preparation & administration:***Initial IV slow bolus:***

Run undiluted over $\geq$ 1 min

Maintenance dose (IV infusion):

Dilute 25,000 units (5ml) up to 50mL of normal saline or dextrose 5%

References:

1. Lexicomp [Mobile version]
2. Pai M., Douketis J.D. (2020). Prevention of VTE in adult orthopedic surgical patients. Available at <https://www.uptodate.com/contents/prevention-of-venous-thromboembolism-in-adult-orthopedic-surgical-patients>.
3. HSgB Farmasi (2021). Guide to Use IV Infusion Heparin Protocol for VTE Treatment.

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COMPARISON¹

	ENOXAPARIN	FONDAPARINUX	HEPARIN
Renal adjustment	Dose adjustment required if CrCl < 30ml/min	Contraindicated if CrCl < 30ml/min	No dosage adjustment
Contra-indications	Immune-mediated HIT within past 100 days or in the presence of circulating antibodies	CrCl < 30ml/min, BW < 50kg (prophylaxis)	Past HIT (with/without thrombosis), severe thrombocytopenia
Pregnancy/Breastfeeding	Yes	Not recommended	Yes
Source	Porcine – No traces of porcine DNA found in the end product ²	Synthetic	Bovine ³
Half-life	4.5–7 hours	17–21 hours	1–2 hours
Antidote	Partially reversible with protamine	NovoSeven®, Prothrombinex® ⁴	Protamine

HIT: Heparin-induced thrombocytopenia.

References:

1. Lexicomp [Mobile version]
2. Sanofi. Clexane outer package.
3. Ain Medicare. Heparinol User Information (2016).
4. Dhakal P, et al. (2016). Reversal of anticoagulation and management of bleeding in patients on anticoagulants. Available at <https://pubmed.ncbi.nlm.nih.gov/27789605/>